

Form Validation

Tasks

You are required to complete a simple form where the user can input a number of fields, simulating a page where a student could register for help on a given unit. The form should ask for 4 separate fields, as shown in the figure below.

The screenshot shows a web browser window with the address bar displaying 'localhost:3000/7-3.html'. The page title is 'Task 7.3C'. The main heading is 'FORM VALIDATION' in large, bold, black letters. Below it is a sub-heading 'Online Form Validation'. A paragraph of text reads: 'Please enter your contact details into the form below to register your interest in receiving support with a given unit. Please make sure you use the correct format when inputting your data.' There are four input fields, each with a label to its left: 'Name:' with a placeholder 'Your name', 'Email:' with a placeholder 'Enter email', 'Deakin Unit:' with a placeholder 'Unit Code', and 'Phone:' with a placeholder 'Phone Number'. At the bottom of the form are two buttons: a blue 'Submit Query' button and a grey 'Reset' button.

Initial Form Prompts

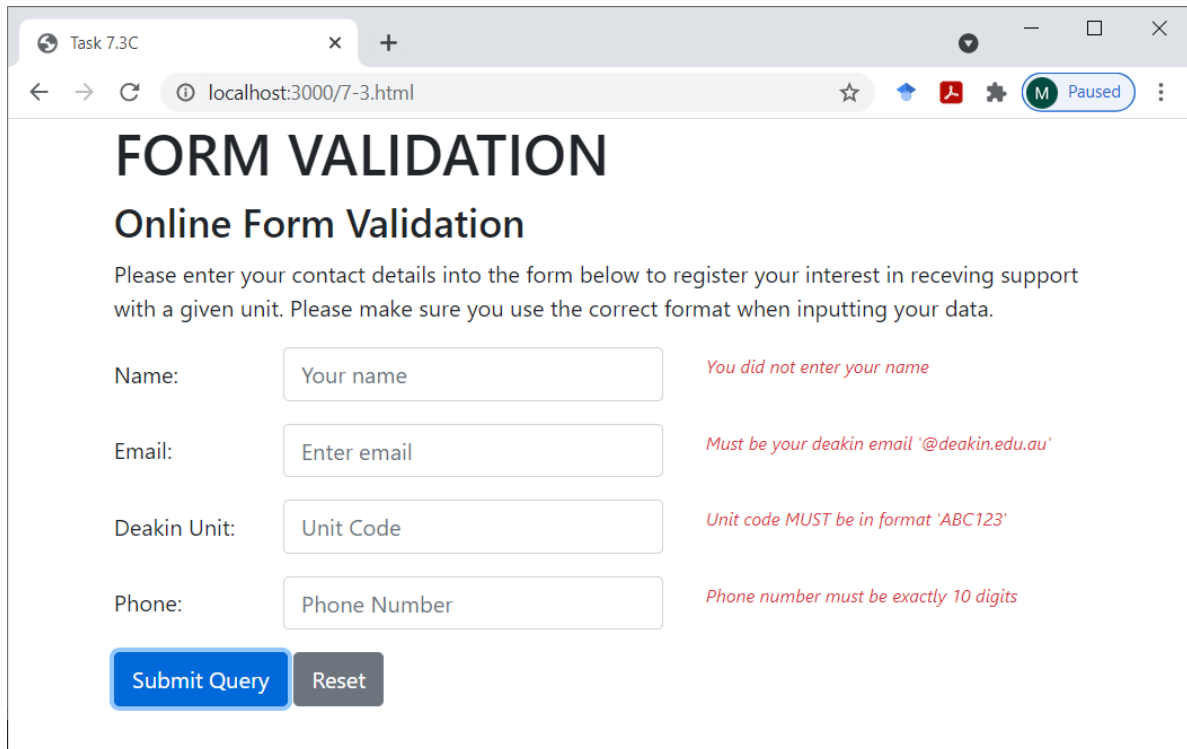
Write a javascript function that allows the data entered by the user to be validated, and where there is an error (the user has input data in an incorrect format) the form should be updated showing the correct format.

The set of:

Form Field	Expected input format rules
Name	This can't be empty. Some input required.
Email	This must be a Deakin email address, i.e., must end in @deakin.edu.au
Deakin Unit Code	Must be 6 characters in size and follow the 3 letters + 3 numbers format (the letters can be capitalised or lowercase)

Form Field	Expected input format rules
Phone Number	Must contain only numbers and be of exactly 10 digits in length

An example of the error messages given to the user are shown in the figure below (where no input is given).



The screenshot shows a web browser window with the title 'Task 7.3C' and the address bar displaying 'localhost:3000/7-3.html'. The page content is titled 'FORM VALIDATION' and 'Online Form Validation'. It includes a paragraph: 'Please enter your contact details into the form below to register your interest in receiving support with a given unit. Please make sure you use the correct format when inputting your data.' Below this, there are four form fields, each with a label, an input box, and a red error message:

- Name: Input box contains 'Your name'. Error message: 'You did not enter your name'.
- Email: Input box contains 'Enter email'. Error message: 'Must be your deakin email '@deakin.edu.au''.
- Deakin Unit: Input box contains 'Unit Code'. Error message: 'Unit code MUST be in format 'ABC123''.
- Phone: Input box contains 'Phone Number'. Error message: 'Phone number must be exactly 10 digits'.

At the bottom of the form, there are two buttons: 'Submit Query' (blue) and 'Reset' (grey).

Error messages on empty fields

In completing this task, you should consider the following requirements:

1. When a field is correct/valid - give this feedback to the user.
2. Show the error message for each of the fields that has invalid input. These may be in zero or more fields, as shown in the example below:

Task 7.3C

localhost:3000/7-3.html

FORM VALIDATION

Online Form Validation

Please enter your contact details into the form below to register your interest in receiving support with a given unit. Please make sure you use the correct format when inputting your data.

Name: Valid

Email: *Must be your deakin email '@deakin.edu.au'*

Deakin Unit: *Unit code MUST be in format 'ABC123'*

Phone: Valid

Invalid email address message

3. When checking if the phone number is valid, if a non-digit is found, display the first instance of this. An example of this is shown below:

Task 7.3C

localhost:3000/7-3.html

FORM VALIDATION

Online Form Validation

Please enter your contact details into the form below to register your interest in receiving support with a given unit. Please make sure you use the correct format when inputting your data.

Name: Valid

Email: Valid

Deakin Unit: Valid

Phone: *Contains character 'w'. Numbers only!*

Phone Number Error

NOTE: Use of Colour of messages is **OPTIONAL**.

Suggested Steps

1. Start with a simple script to check that the fields have not been left blank. If the fields are left blank, an error message should be displayed beside the fields. * And call this by linking to the *onsubmit* event. E.g.,

```

```html
<form onsubmit="validation()">
```

Here the when the form is submitted (when the button is clicked), a javascript function called `validation()` will be called. In this example, as we don't want the actual form to be submitted, so the first line we disable the submit by issuing following command first:

```js
function validation() {
 event.preventDefault();
 ...
}
```

```

2. Extend the script code to check that the email must contain the pattern @deakin.edu.au, i.e., the input mick@gmail.com would be invalid!.
3. Extend the form to also request a *UNIT CODE* and *Phone Number*. The script code should check the following criteria:
 - The *UNIT CODE* must have a length of 6 **and** the first 3 characters must be **letters** and the last 3 characters must be **numbers**
 - The *Phone Number* must be exactly 10 digits in length.
4. Extend the script to check the entered phone number contains only digital numbers. If there is a character in it, the error message must show that character (like the example below).
5. Add a *RESET* button into the form which clears each of the 'error messages'.

Hints

- By naming the *message fields* with *id*'s, these elements in the DOM can be found and manipulated/changed. For example, if the name field was empty, you could: `document.getElementById("namemsg").innerHTML = "<em> You did not enter your name </em>";`
- **OPTIONAL** You can modify class attributes using combinations of: `document.getElementById("namemsg").classList.add("text-danger");` and `document.getElementById("namemsg").classList.remove("text-success");` * https://www.w3schools.com/jsref/prop_element_classlist.asp

- Review the various methods/functions available to work with *JavaScript Regular Expressions* for matching patterns in strings:

- https://www.w3schools.com/js/js_regexp.asp
- https://www.w3schools.com/jsref/jsref_obj_regexp.asp

What will you submit?

You should submit:

- Screenshots (similar to the above examples) of the web page.
- HTML file of the web page with JavaScript.